Appendix: COM105 List of Equations

Units, Physical Quantities, and Vectors

Vector addition

$$\vec{R} = \vec{A} + \vec{B}$$

$$R_x = A_x + B_x$$

$$R_y = A_y + B_y$$

$$R_z = A_z + B_z$$

Unit vector

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$$

Scalar product

$$\vec{A}\cdot\vec{B} = AB\cos\phi = |\vec{A}||\vec{B}|\cos\phi$$

$$\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z$$

Vector product

$$\vec{A}\times\vec{B} = (A_yB_z - A_zB_y)\hat{\imath} + (A_zB_x - A_xB_z)\hat{\jmath} + (A_xB_y - A_yB_x)\hat{k}$$

Motion Along a Straight Line & Motion in 2D or 3D

Average and instantaneous velocity

$$v_{\text{av-}x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

$$v_x = \lim_{\Delta t\to 0}\frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

Average and instantaneous acceleration

$$a_{\text{av-}x} = \frac{\Delta v_x}{\Delta t} = \frac{v_{2x} - v_{1x}}{t_2 - t_1}$$

$$a_x = \lim_{\Delta t\to 0}\frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt}$$

Constant acceleration

$$v_x = v_{0x} + a_xt$$

$$x = x_0 + v_{0x}t + \tfrac{1}{2}a_xt^2$$

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

$$x - x_0 = \tfrac{1}{2}(v_{0x} + v_x)t$$

Varying acceleration

$$v_x = v_{0x} + \int_0^t a_x\,dt$$

$$x = x_0 + \int_0^t v_x\,dt$$

Projectile motion

$$x = (v_0\cos\alpha_0)t$$

$$y = (v_0\sin\alpha_0)t - \tfrac{1}{2}gt^2$$

$$v_x = v_0\cos\alpha_0$$

$$v_y = v_0\sin\alpha_0 - gt$$

Circular motion

$$a_{\text{rad}} = \frac{v^2}{R}$$

$$a_{\text{rad}} = \frac{4\pi^2R}{T^2}$$

Relative velocity

$$v_{P/A\text{-}x} = v_{P/B\text{-}x} + v_{B/A\text{-}x}$$
(relative velocity along a line)

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$$
(relative velocity in space)

Newton's Laws of Motion & Applying Newton's Laws

Friction

Magnitude of kinetic friction force:

$$f_k = \mu_k n$$

Magnitude of static friction force:

$$f_s \le (f_s)_{\max} = \mu_s n$$

Forces in uniform circular motion

$$a_{\text{rad}} = \frac{v^2}{R} = \frac{4\pi^2R}{T^2}$$

Work, Kinetic Energy, Potential Energy, and Energy Conservation

Kinetic energy

$$K = \tfrac{1}{2}mv^2$$

Work-energy theorem

$$W_{\text{tot}} = K_2 - K_1 = \Delta K$$

Work done by varying force

$$W = \int_{x_1}^{x_2} F_x\,dx$$

$$W = \int_{P_1}^{P_2} \vec{F}\cdot d\vec{l}$$

$$= \int_{P_1}^{P_2} F\cos\phi\,dl = \int_{P_1}^{P_2} F_{\parallel}\,dl$$

Power

$$P_{\text{av}} = \frac{\Delta W}{\Delta t}$$

$$P = \lim_{\Delta t\to 0}\frac{\Delta W}{\Delta t} = \frac{dW}{dt}$$

$$P = \vec{F}\cdot\vec{v}$$

Force required to stretch a spring

$$F_x = kx$$

Work done by the gravitational force

$$W_{\text{grav}} = mgy_1 - mgy_2$$
$$= U_{\text{grav},1} - U_{\text{grav},2}$$
$$= -\Delta U_{\text{grav}}$$

Work done by the elastic force

$$W_{\text{el}} = \tfrac{1}{2}kx_1^2 - \tfrac{1}{2}kx_2^2$$
$$= U_{\text{el},1} - U_{\text{el},2} = -\Delta U_{\text{el}}$$

Rotation of Rigid Bodies

Instantaneous angular velocity

$$\omega_z = \lim_{\Delta t\to 0}\frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

Instantaneous angular acceleration

$$\alpha_z = \lim_{\Delta t\to 0}\frac{\Delta\omega_z}{\Delta t} = \frac{d\omega_z}{dt}$$

Constant angular acceleration

$$\theta = \theta_0 + \omega_{0z}t + \tfrac{1}{2}\alpha_z t^2$$

$$\theta - \theta_0 = \tfrac{1}{2}(\omega_{0z} + \omega_z)t$$

$$\omega_z = \omega_{0z} + \alpha_z t$$

$$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0)$$

Linear and angular kinematics

$$v = r\omega$$

$$a_{\tan} = \frac{dv}{dt} = r\frac{d\omega}{dt} = r\alpha$$

$$a_{\text{rad}} = \frac{v^2}{r} = \omega^2 r$$

Moment of inertia

$$I = m_1r_1^2 + m_2r_2^2 + \cdots$$
$$= \sum_i m_i r_i^2$$

Rotational kinetic energy

$$K = \tfrac{1}{2}I\omega^2$$

Parallel-axis theorem

$$I_P = I_{\text{cm}} + Md^2$$

Moment of Inertia of Common Objects

(a) Slender rod, axis through center: $I = \frac{1}{12}ML^2$

(b) Slender rod, axis through one end: $I = \frac{1}{3}ML^2$

(c) Rectangular plate, axis through center: $I = \frac{1}{12}M(a^2 + b^2)$

(d) Thin rectangular plate, axis along edge: $I = \frac{1}{3}Ma^2$

(e) Hollow cylinder: $I = \frac{1}{2}M(R_1^2 + R_2^2)$

(f) Solid cylinder: $I = \frac{1}{2}MR^2$

(g) Thin-walled hollow cylinder: $I = MR^2$

(h) Solid sphere: $I = \frac{2}{5}MR^2$

(i) Thin-walled hollow sphere: $I = \frac{2}{3}MR^2$

Dynamics of Rotational Motion

Torque

$\tau = Fl = rF\sin\phi = F_{\tan}r$

$\vec{\tau} = \vec{r} \times \vec{F}$

$\sum \tau_z = I\alpha_z$

$\sum \vec{\tau} = \frac{d\vec{L}}{dt}$

Combined translational and rotation

$K = \frac{1}{2}Mv_{\text{cm}}^{\ 2} + \frac{1}{2}I_{\text{cm}}\omega^2$

Work done by a torque

$W = \int_{\theta_1}^{\theta_2} \tau_z \, d\theta$

$W = \tau_z(\theta_2 - \theta_1) = \tau_z \Delta\theta$
(constant torque only)

$W_{\text{tot}} = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$

$P = \tau_z\omega_z$

Angular momentum

$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$
(particle)

$\vec{L} = I\vec{\omega}$
(rigid body rotating about axis of symmetry)

Temperature and Heat

Temperature scale

$T_{\text{F}} = \frac{9}{5}T_{\text{C}} + 32°$

$T_{\text{C}} = \frac{5}{9}(T_{\text{F}} - 32°)$

$T_{\text{K}} = T_{\text{C}} + 273.15$

$\frac{T_2}{T_1} = \frac{p_2}{p_1}$

Thermal expansion and stress

$\Delta L = \alpha L_0 \, \Delta T$

$\Delta V = \beta V_0 \, \Delta T$

$\frac{F}{A} = -Y\alpha \, \Delta T$

Heat and temperature/phase changes

$Q = mc \, \Delta T$

$Q = nC \, \Delta T$

$Q = \pm mL$

Conduction

$H = \frac{dQ}{dt} = kA\frac{T_{\text{H}} - T_{\text{C}}}{L}$

Radiation

$H = Ae\sigma T^4$

$H_{\text{net}} = Ae\sigma(T^4 - T_{\text{s}}^4)$

Thermal Properties of Matter

Ideal gas equation

$pV = nRT$

Total mass of substance

$m_{\text{total}} = nM$

Molar mass of substance

$M = N_{\text{A}}m$

Average translational kinetic energy

$K_{\text{tr}} = \frac{3}{2}nRT$

$\frac{1}{2}m(v^2)_{\text{av}} = \frac{3}{2}kT$

RMS speed and mean free path of a gas molecule

$v_{\text{rms}} = \sqrt{(v^2)_{\text{av}}} = \sqrt{\frac{3kT}{m}}$

$\lambda = vt_{\text{mean}} = \frac{V}{4\pi\sqrt{2}\,r^2N}$

Heat capacities

$C_V = \frac{3}{2}R$ (monatomic gas)

$C_V = \frac{5}{2}R$ (diatomic gas)

$C_V = 3R$ (monatomic solid)

Molecular speeds according to Maxwell-Boltzmann distribution

$f(v) = 4\pi\left(\frac{m}{2\pi kT}\right)^{3/2} v^2 e^{-mv^2/2kT}$

The First Law of Thermodynamics

Heat and work

$$W = \int_{V_1}^{V_2} p\,dV$$

$W = p(V_2 - V_1)$ (constant pressure only)

Change in internal energy

$$\Delta U = Q - W$$

$$\Delta U = nC_V(T_2 - T_1)$$

Molar heat capacities

$$C_p = C_V + R$$

Ratio of heat capacities

$$\gamma = \frac{C_p}{C_V}$$

Ideal gas adiabatic process

$$W = nC_V(T_1 - T_2)$$

$$= \frac{C_V}{R}(p_1V_1 - p_2V_2)$$

$$= \frac{1}{\gamma - 1}(p_1V_1 - p_2V_2)$$

; $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$; $p_1V_1^{\gamma} = p_2V_2^{\gamma}$

Heat engines

$$e = \frac{W}{Q_H} = 1 + \frac{Q_C}{Q_H} = 1 - \left|\frac{Q_C}{Q_H}\right|$$

Otto cycle

$$e = 1 - \frac{1}{r^{\gamma-1}}$$

Refrigerators

$$K = \frac{|Q_C|}{|W|} = \frac{|Q_C|}{|Q_H| - |Q_C|}$$

Carnot cycle

$$e_{\text{Carnot}} = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H}$$

$$K_{\text{Carnot}} = \frac{T_C}{T_H - T_C}$$

Entropy

$$\Delta S = \int_1^2 \frac{dQ}{T}$$

(reversible process)

Fundamental Physical Constants

Name	Symbol	Value
Speed of light in vacuum	c	2.99792458×10^8 m/s
Magnitude of charge of electron	e	$1.602176565(35) \times 10^{-19}$ C
Gravitational constant	G	$6.67384(80) \times 10^{-11}$ N·m^2/kg^2
Planck's constant	h	$6.62606957(29) \times 10^{-34}$ J·s
Boltzmann constant	k	$1.3806488(13) \times 10^{-23}$ J/K
Avogadro's number	N_A	$6.02214129(27) \times 10^{23}$ molecules/mol
Gas constant	R	8.3144621(75) J/mol·K
Mass of electron	m_e	$9.10938291(40) \times 10^{-31}$ kg
Mass of proton	m_p	$1.672621777(74) \times 10^{-27}$ kg
Mass of neutron	m_n	$1.674927351(74) \times 10^{-27}$ kg
Magnetic constant	μ_0	$4\pi \times 10^{-7}$ Wb/A·m
Electric constant	$\epsilon_0 = 1/\mu_0c^2$	$8.854187817... \times 10^{-12}$ C^2/N·m^2
	$1/4\pi\epsilon_0$	$8.987551787... \times 10^9$ N·m^2/C^2
Standard atmospheric pressure	1 atm	1.01325×10^5 Pa
Absolute zero	0 K	−273.15°C
Electron volt	1 eV	$1.602176565(35) \times 10^{-19}$ J
Atomic mass unit	1 u	$1.660538921(73) \times 10^{-27}$ kg
Electron rest energy	m_ec^2	0.510998928(11) MeV
Volume of ideal gas (0°C and 1 atm)		22.413968(20) liter/mol
Acceleration due to gravity (standard)	g	9.80665 m/s^2